Measurement of Zero Lift Drag Coefficients for NACA 0012 and NACA 4412 Airfoils Using Wake Profile Analysis

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This experiment determined the zero-lift drag coefficients of the NACA 0012 and NACA 4412 airfoils by analyzing their wake velocity profiles. A constant temperature hot wire anemometer was calibrated using freestream velocities from a pitot-static tube at wind tunnel speeds ranging from 10 Hz to 50 Hz. The resulting calibration followed King's Law, producing a method of conversion between velocity and anemometer voltage. With the airfoils mounted at their respective zero-lift angles of attack, wake profiles were measured approximately one quarter-inch downstream of the trailing edge at 40 Hz. Velocity data was converted, normalized, and integrated using MATLAB to compute drag per unit span and the corresponding drag coefficients. The NACA 0012 yielded a drag coefficient equal to roughly 0.605, while the NACA 4412 yielded roughly 0.0573. The results demonstrated that the cambered 4412 exhibited a thinner wake and slightly lower drag than the symmetric 0012, aligning with expectations including previous lab data. Deviations from published data were attributed primarily to equipment limitations, while mechanical error like vibration noise, and stepper motor inconsistency created local deviations.

I. Nomenclature

α	=	angle of attack, degrees
c	=	chord length, meters
C_D	=	drag coefficient
C_L	=	lift coefficient
D'	=	drag force per unit span, N/m
E	=	measured hot-wire bridge voltage, volts
h	=	vertical extent of wake region, meters
h_1, h_2	=	control-surface heights at inlet and outlet, meters
H_A	=	hot-wire anemometer signal voltage, volts
I	=	electrical current through hot wire
R_W	=	resistance of hot wire
T_W	=	hot-wire temperature, celcius
T_a	=	ambient air temperature, celcius
V_∞	=	freestream velocity, m/s
$u(y)$	=	local velocity within wake profile, m/s
$v(y)$	=	normalized velocity
V_{norm}	=	normalized velocity for plotting
y	=	vertical coordinate within wake, meters
y_{norm}	=	normalized height
ρ	=	air density, kg/m^3
k	=	calibration constant in King's Law
A, B, C	=	empirical calibration constants in King's Law
a, b	=	slope and intercept from log–log calibration fit

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II. Introduction

The objective of this experiment is to determine the zero-lift drag coefficient of two NACA airfoils, 0012 and 4412, by analyzing the velocity change in their wakes. Rather than directly measuring aerodynamic forces, the drag is inferred from the loss of streamwise momentum downstream of the airfoil. This is accomplished by measuring the velocity distribution across the wake using a constant temperature hotwire anemometer mounted on a precision lead screw mechanism.

The hot-wire probe is positioned vertically behind the test airfoil and mounted on a VP9000 stepper motor. The mechanism converts motor revolutions into precise vertical displacements, operating at 400 steps per revolution and 16 revolutions per inch, resulting in a mechanical conversion of 6,400 steps per inch. The probe's position is displayed digitally on the controller, which increments the step count as the probe moves upward through the wake field.

Because the VP9000 internally counts displacement steps, motion during data collection must proceed in one direction (vertically upward) only to prevent positional desynchronization. Prior to each run, the probe is lowered below the wake and the digital counter is zeroed. The system is then advanced manually through short intervals until the probe reaches the top of the wake region, approximately 14,000 steps above the starting position.

The hot wire anemometer is first calibrated to establish a reliable conversion between voltage and velocity. King's Law is the fundamental calibration relationship for a constant-temperature hot wire anemometer. It models how heat loss from a heated wire depends on the velocity of the flow passing over it. As airflow speed increases, the wire loses heat more rapidly through convection, prompting the control circuit to increase electrical power to maintain constant temperature. The relationship between the electrical power (voltage) and the flow velocity is described empirically by King's Law:

$$E^2 = A + BU^n \tag{1}$$

or rearranged in power-law form:

$$U = CE^n \tag{2}$$

where:

- E = measured bridge voltage (V)
- U = air velocity (m/s)
- A, B, C, n = calibration constants

This law arises from convective heat-transfer physics. At steady state, the electrical heating of the wire equals the heat removed by convection:

$$I^2 R_w = hA_s(T_w - T_a) \tag{3}$$

Here, the convective coefficient h increases approximately with U^n, linking voltage and velocity through power law [6]. Instead of solving complex heat-transfer equations for each probe, King's Law provides an empirical shortcut by fitting the exponent n and constants A, B, and C from a small set of calibration measurements. Once the calibration curve is established, measured voltages can be directly converted into accurate velocities within the wake.

In the final part of the experiment , C_D is obtained by applying continuity and streamwise momentum balance to a control volume bounded by control surfaces and streamlines. As standard aerodynamic assumptions are made, the equation for continuity (conservation of mass) holds true and is as follows:

$$\oint_{\mathrm{CS}} \rho V \cdot \hat{n}\, dA = 0 \quad \Rightarrow \quad \int_{\mathrm{CS}_1} \rho V_\infty (-1)\, dy + \int_{\mathrm{CS}_2} \rho\, u(y)(+1)\, dy = 0 \tag{4}$$

$$-\rho V_\infty h_1 + \rho \int_0^{h_2} u(y)\, dy = 0 \quad \Rightarrow \quad h_1 = \frac{1}{V_\infty} \int_0^{h_2} u(y)\, dy \tag{5}$$

Similarly, the momentum balance equation (conservation of momentum) can be rewritten with standard assumptions as

$$\sum F_x = \oint_{\mathrm{CS}} \rho u (V \cdot \hat{n})\, dA = -D' \tag{6}$$

Evaluating the fluxes on each surface, this becomes

$$\int_{\mathrm{CS}_1} \rho u(V\cdot\hat{n})\, dA = \int_0^{h_1} \rho V_\infty(-V_\infty)\, dy = -\rho V_\infty^2 h_1, \tag{7}$$

$$\int_{\mathrm{CS}_2} \rho u(V\cdot\hat{n})\, dA = \int_0^{h_2} \rho\, u^2(y)\, dy. \tag{8}$$

Solving for D' and substituting the equation for h_1 found through continuinty, the final expression for drag can be written as:

$$D' = \rho\int_0^{h_2} u(y)\big(V_\infty - u(y)\big)\, dy \tag{9}$$

Where $u(y)$ is the wake profile as a function of step height. This can be written in the normalized form, where

$$v(y) = \frac{u(y)}{V_\infty},$$

as

$$D' = \rho V_\infty^2 \int_0^{h_2} v(y)\big(1 - v(y)\big)\, dy, \qquad C_D = \frac{D'}{\frac{1}{2}\rho V_\infty^2 c} = \frac{2}{c}\int_0^{h_2} v(y)\big(1 - v(y)\big)\, dy. \tag{10}$$

This formulation directly links measurable wake velocities to aerodynamic drag through conservation of momentum. Here c represents the airfoil chord length, h_2 is the measured wake height, and ρ is the air density. For the purposes of this experiment, air density will be addressed with the uncertainty calculated in a previous lab. Note that the air density will cancel out for all calculations of C_D, but not for D'.

Each local velocity $u(y)$ was obtained from the King's-law calibration applied to the anemometer voltage at each vertical position. The height y was computed from step counts using the traverse lead-screw pitch. For shape comparison, the dimensionless quantities were defined as

$$\mathrm{v}_{\text{norm}} = \frac{U}{U_\infty}, \qquad h_{\text{norm}} = \frac{y}{\max(y)}. \tag{11}$$

As stated, standard aerodynamic assumptions apply: steady, incompressible, and inviscid flow, as well as uniform flow.

III. Procedures

The procedure was divided into three primary parts: first, calibrating the hot wire anemometer, second, collecting data to form the wake profiles, and third, using control volume analysis (momentum balance and continuity) to calculate drag. Testing is conducted using the USE174 wind tunnel at constant temperature and ambient pressure conditions. The vital equipment used in this experiment include a pitot-tube, hot wire anemometer, and step motor system.

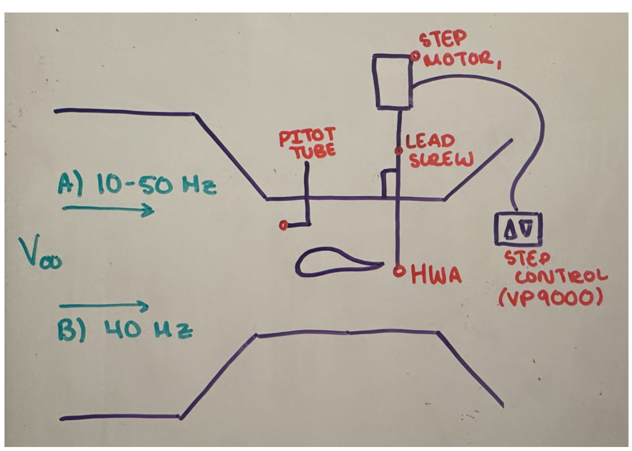


Fig. 1 Detailed Schematic of Test Setup

The hot-wire anemometer was first calibrated to establish a relationship between voltage output and flow velocity. The probe was centered within an empty test section, aligned with the freestream flow. The wind tunnel was operated at fan frequencies ranging from 10 Hz to 50 Hz, in 5 Hz increments, and the corresponding airspeed was measured using a pitot tube connected to a pressure transducer. The measured voltage from the hot wire and the velocity derived from the pitot tube were recorded simultaneously using the lab computer. Data was processed in MATLAB to generate a calibration curve using the power-law form of King's Law (2). This calibration allowed the hot-wire signal to be converted into local velocity during wake profile plotting.

After calibration, the airfoils were individually installed in the test section and mounted at their respective zero-lift angles of attack, $\alpha = 4°$ for NACA 4412, and $\alpha = -°$ for NACA 0012. The hot-wire probe was positioned approximately 1/4 inch downstream of the trailing edge, attached to a VP9000 step motor controller. The step motor provided precise vertical displacements of 400 steps per revolution with a lead screw conversion of 16 revolutions per inch, corresponding to a mechanical conversion of 6,400 steps per inch.

For each test, the probe began at the bottom of the test section and was advanced upward through the wake while maintaining under a constant wind tunnel speed of 40 Hz. The position was zeroed before each run, and motion proceeded only in the upward direction to prevent desynchronization between the controller and motor. Data was collected in specified increments as the probe traversed across the wake. The following step sizes were used for increased measurement fidelity near the wake:

NACA 0012	Step Size (steps)	Range
	400	0 – 3200
	200	3200 – 4200
	100	4200 – 5400
	200	5400 – 6400
	400	6400 – 10,000
	1000	10,000 – 14,000

NACA 4412	Step Size (steps)	Range
	400	0 – 4000
	200	4000 – 5200
	100	5200 – 6000
	200	6000 – 7200
	400	7200 – 10,000
	1000	10,000–14,000

Once all wake velocity data was collected, the measured voltages were converted into velocities using the previously derived calibration curve. Velocities and heights were normalized to create dimensionless profiles, allowing direct comparison between airfoils. The normalized data was integrated numerically in MATLAB to determine the drag per unit span D' and corresponding drag coefficients C_D. This was done through the control volume analysis detailed in Section II. The following diagram further illustrates the control volume chosen, and derivation of the equations themselves.

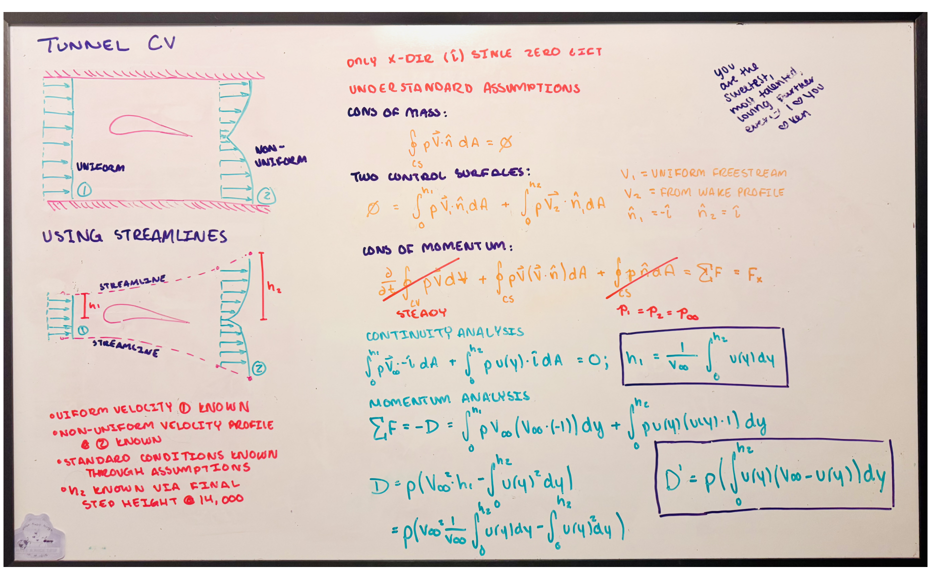


Fig. 2 Control Volume Analysis and Flow for Processing

Once again, all data processing, plotting, and control volume integrations were performed in MATLAB. The results were compared against theoretical expectations from Abbott data as well as prior lab results.

IV. Results

The hot wire anemometer was calibrated first to establish an accurate conversion between the measured voltage and its respective velocity. This was done by exposing the hot wire to the freestream air within the wind tunnel, recording true wind speed through a pitot tube as well as the anemometer voltage. Data was recorded for 9 different wind speeds, or wind tunnel settings, in increments of 5hz between 10Hz and 50Hz. All recorded data was processed in MATLAB to produce the following graph:

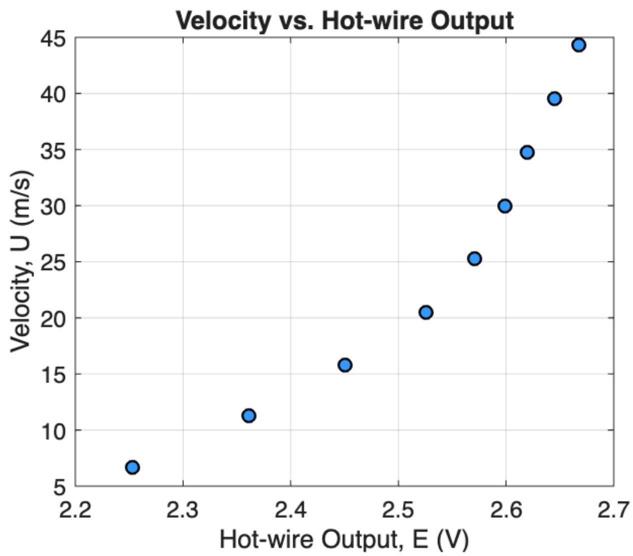


Fig. 3 Velocity vs. Hot-wire Output

The plot shows that as velocity (U) increases, the hot wire output voltage (E) also increases. The relationship is non-linear, showing that faster air cools down the hot wire more via convection. This requires more power (higher voltage) to be delivered to the anemometer to keep the constant temperature.

King's Law was used model how the heat loss from the anemometer relates to the velocity of the air passing over it. Rearranging eqs. (2), the logarithmic expression for King's Law is produced:

$$\log(U) = n\log(E) + \log(C). \tag{12}$$

This was the relationship used for the calibration curve, where the slope of the linear fit (n) and the intercept ($\log(C)$ were returned to eqs. (2) to create the final calibration equation

$$U = 9.05 \times 10^{-4} E^{10.93}.$$

This was the equation used to convert any hot wire voltage into usable velocity. The MATLAB processing done to achieve these values will be included in the appendix, and the graph produced is as follows:

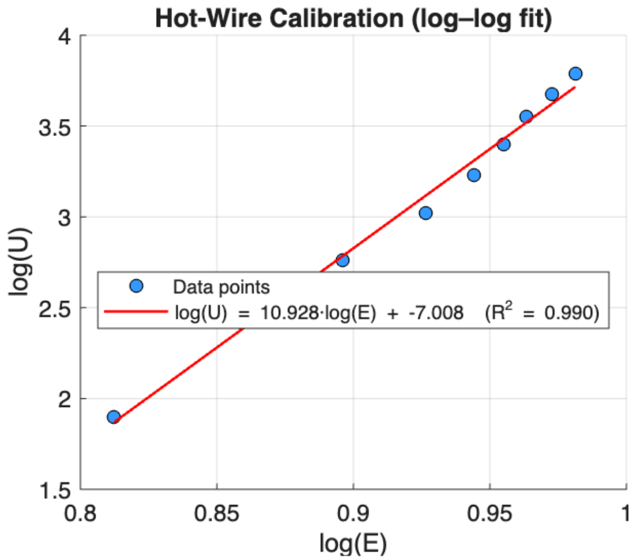
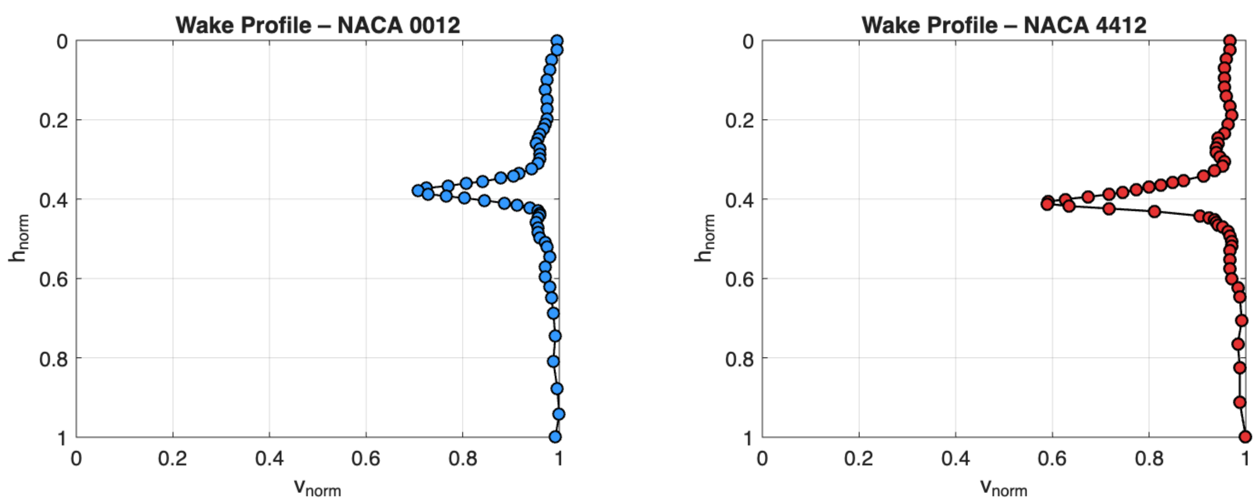


Fig. 4 Hot-Wire Calibration Curve

Note that R^2, the coefficient of determination, was extremely close to 1, indicating the regression line to be a very accurate representation of the data.

The hot-wire calibration was then used to convert the measured voltages behind the airfoil into velocity profiles, normalizing both height and velocity to create a dimensionless curve revealing the shape of the wake. The normalization for each dimension was done through eqs. (11), and the respective conversions for each was used. For both airfoils, all data was processed, converted, and plotted in MATLAB. The code will be in the appendix, and the resulting figures are as follows:



(a) Wake Profile for NACA 0012 **(b) Wake Profile for NACA 4412**

Fig. 5 Comparison of wake profiles for NACA 0012 and NACA 4412.

Note that data collection, in terms of the steps, was not exact. By nature, some measurements were tens to hundreds of steps off from the intended hot wire position. The step sizes were chosen so that there was higher fidelity in positions where the velocity changes the most. This explains the varying gaps in data shown on each wake profile.

In Fig. 5, the NACA 0012 shows a wide region of reduced velocity behind the airfoil. In contrast, the NACA 4412 shows a large spike, but much faster recovery than 0012. It should be noted that 4412 shows significantly more turbulence than 0012, as shown by the waviness of the wake profile. Using the wake velocity profiles shown in Fig. 5, the drag per unit span (D') and the corresponding drag coefficients (C_D) were computed through conservation laws (10). Applying the relationships derived in Section II, and automating the calculations using a MATLAB code referenced in the appendix, the following results were obtained.

$$D'_{0012} = 4.0997 \pm 0.0141 \text{ N/m}, \qquad C_{D,0012} = 0.06054$$

$$D'_{4412} = 3.8406 \pm 0.0132 \text{ N/m}, \qquad C_{D,4412} = 0.05734$$

Both values are consistent with expectations, where the cambered NACA 4412 airfoil exhibits slightly less drag compared to the symmetric NACA 0012. This reduction in drag corresponds to the previously discussed thinner wake observed in Fig. 5(b). The results show that the wake profile effectively captures differences in aerodynamic values between airfoils. The small difference in C_D values reflects the sensitivity of the data collection process.

In a previous experiment, data was collected to build various plots displaying aerodynamic coefficients and how they relate to different angles of attack. In the experiment, the value for C_l was never cited explicitly, but the plot that includes C_l shows that the value for NACA 0012 at $\alpha = 0°$ was somewhere between $0.04 \leq C_l \leq 0.06$, while the NACA 4412 value for C_l at $\alpha = 4°$ was somewhere between $0.06 \leq C_l \leq 0.09$ [7]. This validates the data collected in this experiment, as the values achieved, are within the expected range shown in the previous lab.

Additionally, Abbott data from *Theory of Wing Sections* cites that at zero-lift angle of attack, the NACA 0012 airfoil should have a C_D of roughly 0.006 to 0.009. For a NACA 4412 airfoil operated at the zero-lift angle of attack, the range of C_D should be around 0.008-0.012 [8]. These are orders of magnitude off from the values achieve in this lab. Though there are many reasons why this could be, it is likely from a vast difference in lab equipment, method of calculation, conditions, and scale.

V. Conclusion

The experimental determination of drag coefficients for the NACA 0012 and 4412 airfoils successfully demonstrated the use of wake profile measurements to evaluate aerodynamic performance. The results showed that the cambered NACA 4412 airfoil produced a thinner wake and slightly lower drag coefficient. This outcome aligns with aerodynamic expectations, as camber generally reduces drag at the zero-lift angle of attack. Although the experimental values were higher than published Abbott data, the relative trends between the two airfoils were consistent, validating the experimental approach.

There are many sources of error: machine uncertainty, calculated density uncertainty, negated values from assumptions, and more. However, the most impactful source of error was likely the hot-wire a stepper mechanism in tandem. The hot-wire is susceptible to vibrations from the wind tunnel, which make the airspeed relative to the vibrations the hot-wire is experiencing. This is similar to waving an object in the air to cool it off. Additionally, there is a natural temperature loss from the hot-wire being cooled down by the air around it. This is what would happen if there was no air flow. Additionally, there are data collection limits present, namely in how the stepper motor was used to collect data at various points. In the ideal case, the experiment is conducted with data collection at every single step, as it would provide the most amount of data and thereby the most accurate results. Of course, this would be extremely arduous, hence why step intervals were taken in a high fidelity where velocity changes the most. On Fig. 5 (b), there is a noticeable gap in data, demonstrating the present impact of this error.

Despite these limitations, the experiment effectively demonstrated the physical principles connecting the uses for hot-wire anemometers, control volume analysis, and the aerodynamic properties. The wake profile method provided a practical and non-intrusive means of determining drag. Future improvements could include finer step resolution, and active vibration isolation.

Appendix A

Table 2 Table A-1: Density and Uncertainty

Property	Symbol	Value	Units	Uncertainty
Air Density	ρ	1.1069	kg/m^3	±0.00381

Note: The air density and associated uncertainty were obtained from Lab 1 using the pressure–temperature method. This value was used in all subsequent drag calculations for both airfoils.

Table 3 Table A-2: Calibration Data

Air Temperature (°C)	Dynamic Pressure (Pa)	Hotwire Output (V)	Air Velocity (m/s)
29.787	24.841	2.253	6.690
30.023	69.977	2.361	11.233
30.175	138.666	2.450	15.817
30.350	232.182	2.526	20.473
30.586	353.311	2.571	25.265
30.856	496.904	2.599	29.976
31.262	668.370	2.620	34.788
31.825	863.104	2.645	39.569
32.377	1079.210	2.668	44.286

Table 4 Table A-3: NACA 0012 Normalized Velocity Data

Step	y (m)	$u(y)$ (m/s)	$v = u/U_\infty$	h_{norm}
400	0.0015875	34.2069	0.9779	0.0248
800	0.0031750	33.7817	0.9657	0.0497
1200	0.0047625	33.6411	0.9617	0.0745
1600	0.0063500	33.5009	0.9577	0.0994
2000	0.0079375	33.3613	0.9537	0.1242
2400	0.0095250	33.5009	0.9577	0.1490
2800	0.0111125	33.5009	0.9577	0.1739
3200	0.0127000	33.5009	0.9577	0.1987
3405	0.0135136	33.3613	0.9537	0.2144
3600	0.0142875	33.2223	0.9497	0.2236
3820	0.0151606	32.9457	0.9418	0.2372
4000	0.0158750	32.8082	0.9379	0.2484
4200	0.0166875	32.6712	0.9339	0.2601
4408	0.0174943	32.9457	0.9418	0.2737
4624	0.0183515	32.9457	0.9418	0.2871
4801	0.0190539	32.9457	0.9418	0.2981
5000	0.0198438	32.8082	0.9379	0.3105
5211	0.0206812	32.3988	0.9262	0.3236

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Step	y (m)	$u(y)$ (m/s)	$v = u/U_\infty$	h_{norm}
5409	0.0214669	31.4616	0.8994	0.3359
5501	0.0218321	31.0676	0.8881	0.3416
5600	0.0222225	30.1655	0.8624	0.3477
5724	0.0227171	28.9180	0.8267	0.3554
5807	0.0230465	27.7176	0.7924	0.3606
5900	0.0234156	26.4496	0.7561	0.3664
6000	0.0238125	24.9119	0.7122	0.3728
6100	0.0242094	24.2783	0.6941	0.3788
6237	0.0247531	25.0191	0.7152	0.3873
6325	0.0251023	26.3369	0.7529	0.3928
6400	0.0254000	27.6001	0.7890	0.3974
6500	0.0257969	29.0406	0.8302	0.4036
6600	0.0261938	30.4208	0.8697	0.4098
6700	0.0265906	31.3298	0.8956	0.4160
6803	0.0269999	32.2634	0.9223	0.4224
6920	0.0274638	32.8082	0.9379	0.4297
7003	0.0277932	32.9457	0.9418	0.4349
7100	0.0281781	32.9457	0.9418	0.4409
7200	0.0285750	32.8082	0.9379	0.4471
7400	0.0293688	32.6712	0.9339	0.4595
7601	0.0301665	32.8082	0.9379	0.4719
7810	0.0309959	32.8082	0.9379	0.4849
8001	0.0317539	32.9457	0.9418	0.4968
8200	0.0325438	33.3613	0.9537	0.5092
8400	0.0333338	33.5009	0.9577	0.5216
8800	0.0349250	33.6411	0.9617	0.5464
9200	0.0365125	33.3613	0.9537	0.5713
9602	0.0381079	33.3613	0.9537	0.5962
10004	0.0397034	33.6411	0.9617	0.6212
10429	0.0413901	33.7817	0.9657	0.6476
11063	0.0439063	33.9229	0.9698	0.6870
12012	0.0476726	34.0646	0.9738	0.7459
13010	0.0516334	33.9229	0.9698	0.8079
14132	0.0560864	34.2069	0.9779	0.8775
15152	0.0601345	34.3497	0.9820	0.9409
16104	0.0639128	34.0646	0.9738	1.0000

END OF DATA

Table 5 Table A-4: NACA 4412 Normalized Velocity Data

Step	y (m)	$u(y)$ (m/s)	$v = u/U_\infty$	h_{norm}
400	0.0015875	33.6411	0.9671	0.0235
800	0.0031750	33.3613	0.9590	0.0471
1200	0.0047625	33.2225	0.9550	0.0706
1609	0.0063857	33.2225	0.9550	0.0946
2007	0.0079653	33.2225	0.9550	0.1181
2400	0.0095250	33.3613	0.9590	0.1411
2803	0.0111244	33.6411	0.9671	0.1648
3200	0.0127000	33.7817	0.9711	0.1882
3600	0.0142875	33.5009	0.9630	0.2117
4000	0.0158750	33.2225	0.9550	0.2353
4200	0.0166875	32.8082	0.9432	0.2471
4400	0.0174625	32.8082	0.9432	0.2588
4600	0.0182625	32.6712	0.9392	0.2706
4800	0.0190500	32.6712	0.9392	0.2823
5000	0.0198438	32.9457	0.9471	0.2941
5200	0.0206375	33.2225	0.9550	0.3059
5400	0.0214313	33.0837	0.9510	0.3176
5600	0.0222225	32.5348	0.9353	0.3294
5800	0.0230188	31.7269	0.9120	0.3412
6008	0.0238443	30.2929	0.8708	0.3534
6100	0.0242094	29.5358	0.8490	0.3588
6200	0.0246063	28.6742	0.8243	0.3647
6300	0.0250031	27.8356	0.8002	0.3706
6402	0.0254079	26.9044	0.7734	0.3766
6505	0.0258176	25.8908	0.7443	0.3826
6600	0.0261938	24.9119	0.7161	0.3882
6700	0.0265906	23.4560	0.6743	0.3941
6809	0.0270232	21.7923	0.6264	0.4005
6907	0.0274122	20.5931	0.5919	0.4063
7004	0.0277911	20.5034	0.5894	0.4120
7100	0.0281781	22.0778	0.6347	0.4176
7204	0.0285909	24.9119	0.7161	0.4237
7325	0.0290711	28.1922	0.8104	0.4309
7519	0.0298410	31.4616	0.9044	0.4423
7600	0.0301625	32.1285	0.9236	0.4471
7700	0.0305594	32.5348	0.9353	0.4529
7800	0.0309656	32.6712	0.9392	0.4588
7902	0.0313611	32.8082	0.9431	0.4648
8000	0.0317500	33.0837	0.9510	0.4706
8200	0.0325438	33.5009	0.9630	0.4823
8400	0.0333375	33.6411	0.9671	0.4941

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Table A-4: NACA 4412 Normalized Velocity Data (continued)

Step	y (m)	$u(y)$ (m/s)	$v = u/U_\infty$	h_{norm}
8600	0.0341313	33.7817	0.9711	0.5059
8800	0.0349250	33.7817	0.9711	0.5176
9000	0.0357188	33.6411	0.9671	0.5294
9400	0.0373063	33.6411	0.9671	0.5529
9800	0.0388938	33.6411	0.9671	0.5765
10201	0.0404852	33.7817	0.9711	0.6001
10604	0.0420846	34.2069	0.9833	0.6238
11000	0.0436563	34.3497	0.9874	0.6471
12008	0.0476675	34.4931	0.9915	0.7064
13001	0.0515978	34.2069	0.9833	0.7647
14006	0.0555861	34.3497	0.9874	0.8239
15502	0.0615236	34.3497	0.9874	0.9119
17000	0.0674688	34.7810	0.9999	1.0000

END OF DATA

Table 6 Table A-5: Experimental Drag Results

Airfoil	U_∞ (m/s)	D' (N/m)	Uncertainty (N/m)	C_D
NACA 0012	34.98	4.0997	± 0.0141	0.06054
NACA 4412	34.79	3.8406	± 0.0132	0.05734

Appendix B

(ii) Derivation of drag calculation

With steady, incompressible flow and streamline boundaries, the continuity equation over the control volume reduces to fluxes through the two control surfaces only:

$$\int_{\mathrm{CS}} \rho\, V \cdot \hat{n}\, dA \;=\; 0 \quad\Rightarrow\quad \int_{\mathrm{CS}_1} \rho\, V_1 \cdot \hat{n}_1\, dA \;+\; \int_{\mathrm{CS}_2} \rho\, V_2 \cdot \hat{n}_2\, dA = 0.$$

On CS_1 the flow is uniform, $V_1 = V_\infty \hat{\imath}$ and $\hat{n}_1 = -\hat{\imath}$, so

$$\int_{\mathrm{CS}_1} \rho\, V_1 \cdot \hat{n}_1\, dA = \int_0^{h_1} \rho\, V_\infty (-1)\, dy = -\rho V_\infty h_1.$$

On CS_2 the measured wake has $V_2 = u(y)\hat{\imath}$ and $\hat{n}_2 = +\hat{\imath}$, giving

$$\int_{\mathrm{CS}_2} \rho\, V_2 \cdot \hat{n}_2\, dA = \int_0^{h_2} \rho\, u(y)\, (+1)\, dy = \rho \int_0^{h_2} u(y)\, dy.$$

Therefore,

$$-\rho V_\infty h_1 \;+\; \rho \int_0^{h_2} u(y)\, dy = 0 \qquad\Rightarrow\qquad h_1 \;=\; \frac{1}{V_\infty} \int_0^{h_2} u(y)\, dy\,.$$

This relation closes the control-volume system by expressing the unknown inlet height h_1 in terms of the measured wake profile and the freestream. The conservation of momentum equation is as follows:

$$\sum F_x = \int_{\mathrm{CS}} \rho\, u\, (V \cdot \hat{n})\, dA.$$

By standard assumptions, the integral term for unsteady flow is negated, as well as the pressure forces integral since $p_1 = p_2 = p_\infty$. There are no forces in the x-direction other than the drag induced by the airfoil, D'. Thereby, $\sum F_x = -D'$. Evaluating the flux terms,

$$\int_{\mathrm{CS}_1} \rho\, u(V \cdot \hat{n}_1)\, dA = \int_0^{h_1} \rho\, V_\infty(-V_\infty)\, dy = -\rho V_\infty^2 h_1,$$

$$\int_{\mathrm{CS}_2} \rho\, u(V \cdot \hat{n}_2)\, dA = \int_0^{h_2} \rho\, u(y)\, u(y)\, dy = \rho \int_0^{h_2} u(y)^2\, dy.$$

Hence,

$$-D' \;=\; \rho \int_0^{h_2} u(y)^2\, dy \;-\; \rho V_\infty^2 h_1 \qquad\Rightarrow\qquad D' \;=\; \rho\Big(V_\infty^2 h_1 - \int_0^{h_2} u^2\, dy\Big).$$

Plugging in the known h_1, derived from the conservation of mass equation, the equation becomes:

$$D' = \rho \left[V_\infty \int_0^{h_2} u\, dy - \int_0^{h_2} u^2\, dy \right] = \rho \int_0^{h_2} u(y)\, (V_\infty - u(y))\, dy.$$

Defining the normalized velocity as $v(y) = u(y)/V_\infty$, the equation for D' becomes

$$D' = \rho V_\infty^2 \int_0^{h_2} v(1 - v)\, dy, \qquad ,$$

and substituting the known equation for C_D as a function of D', the final form for the drag calculations is achieved:

$$C_D \;=\; \frac{D'}{\frac{1}{2}\rho V_\infty^2 c} \;=\; \frac{2}{c} \int_0^{h_2} v(y)\, \big(1 - v(y)\big)\, dy\,.$$

This is the expression used to compute C_D from the measured wake profile, with y obtained from the stepper conversion and h_2 equal to the final traverse height (1400). Note that although air density uncertainty is required, the term is ultimately canceled in C_D calculations. For this reason, all drag calculations include both C_D and D'.

(i) Sample Calculation — NACA 0012

Given/Constants:

$$C = 9.05 \times 10^{-4}, \qquad n = 10.93, \qquad \rho = 1.1069 \text{ kg/m}^3,$$
$$c = 0.100 \text{ m}, \qquad \text{step size } \Delta y = \frac{0.0254}{6400} = 3.96875 \times 10^{-6} \text{ m/step}.$$

Data processing:

Note: The following steps are reflective of the calculations done by the MATLAB code.

1) Convert steps s_i to vertical heights:

$$y = (s - s_1)\,\Delta y, \qquad \text{(for reference, } h_2 \approx 14000\,\Delta y = 0.05556 \text{ m)}.$$

2) Convert hot–wire voltages E_i to local velocities (using King's law):

$$U = C\,E^{n}.$$

3) Freestream speed from dynamic pressure q:

$$U_\infty = \sqrt{\frac{2\,q}{\rho}} = \sqrt{\frac{2(677.2 \text{ Pa})}{1.1069}} = 34.98 \text{ m/s}.$$

4) Normalize the wake:

$$v(y) \equiv \frac{u(y)}{U_\infty}.$$

5) Momentum integral using trapezoidal rule over the measured span of y:

$$\int_0^{h_2} v(y)(1 - v(y))\,dy \;=\; \text{trapz}\big(y,\; v\,(1 - v)\big) = 0.0030269 \text{ m}.$$

The final result for C_D is as follows:

$$C_D = \frac{2}{c}\int_0^{h_2} v(1 - v)\,dy = \frac{2}{0.100}\,(0.0030269) = 0.06054.$$

The final result for D' is as follows:

$$D' = \rho U_\infty^2 \int_0^{h_2} v(1 - v)\,dy = (1.1069)(34.98)^2(0.0030269) = 4.0997 \text{ N/m},$$

as well as the uncertainty, once again only present in D',

$$\frac{\delta D'}{D'} = \frac{\delta \rho}{\rho} = \frac{0.00381}{1.1069} = 0.00344\ (0.344\%), \qquad \Rightarrow \qquad D' = 4.0997 \pm 0.0141 \text{ N/m}\,.$$

Appendix C

A. Hot-wire Data Visualization and Calibration

```
    clear; clc; close all
% hot wire calibration
fname = 'Fri_A_1400_calibration.txt';

T = readtable(fname,'VariableNamingRule','preserve');
E = T{:,3}; % hotwire output in volts
U = T{:,4}; % velocity

% plot U vs. HWA
figure(1); clf
plot(E,U,'ko','MarkerFaceColor',[0.2 0.6 1],'LineWidth',1.1)
grid on
xlabel('Hot-wire Output, E (V)')
ylabel('Velocity, U (m/s)')
title('Velocity vs. Hot-wire Output','FontSize',13)
set(gca,'FontSize',11,'Box','on')
set(gcf,'Color','w')

% plot log/log for kings law linearization
x = log(E); y = log(U); % natural logs
p = polyfit(x,y,1); % y = a*x + b
a = p(1); b = p(2);
Ufit = exp(b)*E.^a; % U = exp(b) * E^a

% R^2 (on loglog space)
yhat = polyval(p,x);
R2 = 1 - sum((y - yhat).^2)/sum((y - mean(y)).^2);

figure(2); clf
hold on
plot(x,y,'ko','MarkerFaceColor',[0.2 0.6 1],'DisplayName','Data points')
plot(x,yhat,'r-','LineWidth',1.2, ...
'DisplayName',sprintf('log(U) = %.3flog(E) + %.3f  (R^2 = %.3f)',a,b,R2))
hold off
grid on
xlabel('log(E)'); ylabel('log(U)')
title('Hot-Wire Calibration (loglog fit)','FontSize',13)
legend('Location','northwest','FontSize',9)
set(gcf,'Color','w'); set(gca,'FontSize',11)

% printing summary
fprintf('King''s-law form (power law fit):  U = C * E^{n}\n');
fprintf('  n = %.6f\n',a);
fprintf('  C = exp(b) = %.6f  (units: m/sV^{-n})\n',exp(b));
fprintf('  R^2 (loglog fit) = %.6f\n',R2);
```

B. Airfoil Wake Profiles

Note: This code was used to create the wake profiles for both for NACA 0012 and NACA 4412. The only thing changed was the .txt file, and data color to differential the two plots.

```
clc; clear; close all
fname = 'Fri_A_1400_0012.txt';
T = readtable(fname,'VariableNamingRule','preserve');

E = T{:,3}; % hot wire voltage
steps = T{:,6}; % step position
U = 9.05e-4 * E.^10.93; % calibration from part A

% normalizing
U_inf = max(U);
U_norm = U ./ U_inf;
y = steps / 6400;
y_norm = y ./ max(y);

% plotting
figure; clf
plot(U_norm, y_norm, 'ko-', 'MarkerFaceColor',[0.2 0.6 1], 'LineWidth',1)
grid on
xlabel('v_{norm}','FontSize',12)
ylabel('h_{norm}','FontSize',12)
title('Wake␣Profile␣␣NACA␣0012','FontSize',13)

xlim([0 1])
ylim([0 1])
set(gca,'YDir','reverse')
set(gca,'FontSize',11,'Box','on')
set(gcf,'Color','w')
```

C. Drag Calculation and Analysis

Note: This code was used calculate the drag for both for NACA 0012 and NACA 4412. The only thing changed was the .txt file.

```
clc; clear; close all
% constants
C = 9.05e-4; % Kings law constant
n = 10.93; % Kings law exponent
rho = 1.1069; % air density [kg/m^3]
c = 0.100; % chord from lab 2 info [m]
dy_per_step = 0.0254 / 6400; % step size [m] (16 rev/in * 400 steps/rev)

fname = 'Fri_A_1400_0012.txt';
T = readtable(fname,'FileType','text');
steps = T{:,6};
E = T{:,3};
q = T{:,2};

% step conversion for a height
y = (steps - steps(1)) * dy_per_step;
% velocity conversion from kings law
u = C .* (E.^n);
% freestream velocity from dynamic pressure
Uinf = sqrt(2 * mean(q) / rho);
% normalize that
v = u / Uinf;
% computing drag coeffcient from drag per eqs.
Cd = (2 / c) * trapz(y, v .* (1 - v));
% throwing drap per eqs. in there
Dprime = rho * trapz(y, u .* (Uinf - u));
% uncertainty in air density
rho_unc = 0.00381; % kg/m^3
Dprime_unc = Dprime * (rho_unc / rho);

% results
fprintf('File:␣%s\n', fname);
fprintf('Freestream␣velocity:␣%.2f␣m/s\n', Uinf);
fprintf('Drag␣per␣span␣D''␣=␣%.4f␣␣%.4f␣N/m\n', Dprime, Dprime_unc);
fprintf('Drag␣coefficient␣C_D␣=␣%.5f\n', Cd);
```

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